Master: Bases de Données Fall, 2026

TP 3 (universityDB-data)

1 Setting up PostgreSQL (if not done already)

USING USPN LABS DATABASE

- Login to `linux` with your user account.
- Type the following command to start the `postgreSQL prompt`:

  ```
  psql -h postgres -U m1info_xx
  ```

  where `xx` is your account number between 1 and 90. The password is `M1INFO_xx`, with the same value of `xx` as your account number.
- You can see the list of tables present by typing :

  ```
  \d
  ```

2 Exercises

Now write the SQL code, and then run to verify, the following queries. The table after each question gives the result that you should get.

a.) Find the name of the teacher with the 4-th highest salary. Assume the salaries are distinct. Write this query using **WITH**.

Trouvez le nom de l'enseignant ayant le 4e salaire le plus élevé. Supposez que les salaires sont distincts. Écrivez cette requête en utilisant **WITH**.

name
Gold

Solution:

```
WITH T (name, nb) AS
     (SELECT t1.name, count(*)
      FROM teacher AS t1, teacher AS t2
      WHERE t1.salary < t2.salary
      GROUP BY t1.name)
SELECT name
FROM T
WHERE nb = 3;
```

b.) Find the `ids` and `names` of all students who have not taken any course before 2010.

Trouvez les `ids` et `names` de tous les étudiants qui n'ont suivi aucun cours avant 2010.

id	*name*
55739	Sanchez
19991	Brandt
23121	Chavez
70557	Snow

Solution:

```
(SELECT student.id, student.name
FROM student)
EXCEPT
(SELECT student.id, student.name
FROM student, takes
WHERE takes.id = student.id and takes.year < 2010)
```

c.) Find students who have taken all courses offered by their own department.

Trouvez les étudiants qui ont suivi tous les cours offerts par leur propre département.

id	*name*
19991	Brandt
23121	Chavez
44553	Peltier
55739	Sanchez
76653	Aoi

Solution:

```
SELECT s.ID, s.name
FROM student AS s
WHERE NOT EXISTS (
    SELECT c.course_id
    FROM course AS c
    WHERE c.dept_name = s.dept_name
    AND c.course_id NOT IN (
        SELECT t.course_id
        FROM takes AS t
        WHERE t.ID = s.ID
    )
);
```

d.) Find students who share exactly the same set of course_ids (their sections/semester/year can be different!) as at least one other student. Use **EXISTS** and **EXCEPT**.

Trouvez les étudiants qui partagent exactement le même ensemble de course_ids (leurs sections/semestre/année peuvent être différentes !) avec au moins un autre étudiant. Utilisez **EXISTS** et **EXCEPT**.

id	*name*
45678	Levy
76543	Brown

```
SELECT s1.id, s1.name
FROM student AS s1
WHERE EXISTS (
    SELECT s2.id
    FROM student s2
    WHERE s2.id != s1.id
    AND NOT EXISTS (
        SELECT t1.course_id
        FROM takes AS t1
        WHERE t1.id = s1.id
        EXCEPT
        SELECT t2.course_id
        FROM takes AS t2
        WHERE t2.id = s2.id
    )
    AND NOT EXISTS (
        SELECT t3.course_id
        FROM takes AS t3
        WHERE t3.id = s2.id
        EXCEPT
        SELECT t4.course_id
        FROM takes AS t4
        WHERE t4.id = s1.id
    )
);
```

e.) Find teachers who have taught courses with all different credit values available in their department (using **NOT EXISTS** and **EXCEPT**).

Trouvez les enseignants qui ont enseigné des cours couvrant toutes les valeurs de crédits différentes disponibles dans leur département (en utilisant **NOT EXISTS** et **EXCEPT**).

id	*name*
10101	Srinivasan
12121	Wu
15151	Mozart
22222	Einstein
32343	El Said
45565	Katz
83821	Brandt
98345	Kim

```
SELECT t.id, t.name
FROM teacher AS t
WHERE NOT EXISTS (
    SELECT DISTINCT c.credits
    FROM course AS c
    WHERE c.dept_name = t.dept_name
    EXCEPT
    SELECT DISTINCT c2.credits
    FROM teaches AS te
    JOIN course c2 ON te.course_id = c2.course_id
    WHERE te.id = t.id
);
```

f.) List all pairs of students who have taken most classes together. Do not use products, use only **JOINS**.

Listez toutes les paires d'étudiants ayant suivi ensemble le plus grand nombre de cours. N'utilisez pas de produits cartésiens, utilisez uniquement des **JOINS**.

name	*name*
Shankar	Zhang
Shankar	Williams
Bourikas	Shankar

```
SELECT A.name, B.name
FROM (student NATURAL INNER JOIN takes) as A
       CROSS JOIN
     (student NATURAL INNER JOIN takes) as B
WHERE A.course_id = B.course_id AND A.sec_id = B.sec_id
      AND A.semester = B.semester AND A.year = B.year
      AND A.id <> B.id AND A.name < B.name
GROUP BY A.name, A.id, B.name, B.id
HAVING count(*) >= ALL(
   SELECT count(*)
   FROM (student NATURAL INNER JOIN takes) as A
          CROSS JOIN
        (student NATURAL INNER JOIN takes) as B
   WHERE A.course_id = B.course_id AND A.sec_id = B.sec_id
        AND A.semester = B.semester AND A.year = B.year
        AND A.id <> B.id AND A.name < B.name
  GROUP BY A.id, B.id);
```

g.) Give the number of 'A' grades given by **each** teacher. Do not use products, use only **JOINS**.

Donnez le nombre de notes 'A' attribuées par **chaque** enseignant. N'utilisez pas de produits cartésiens, utilisez uniquement des **JOINS**.

name	*count*
Srinivasan	4
Brandt	2
Crick	1
Califieri	0
Einstein	0
Wu	0
Kim	0
Mozart	0
El Said	0
Katz	0
Singh	0
Gold	0

```
WITH mytakes (id, course_id, sec_id, semester, year, grade) AS
   (SELECT id, course_id, sec_id, semester, year, grade
    FROM takes
    WHERE grade = 'A')

SELECT teacher.name, count(course_id)
FROM (mytakes INNER JOIN teaches USING (course_id, sec_id,
     semester, year))
       RIGHT OUTER JOIN teacher ON teaches.id = teacher.id
GROUP BY teacher.name, teacher.id
ORDER BY count(course_id) DESC;
```

Tables

course_id	*title*	*dept_name*	*credits*
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4

(a) course

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

(b) teacher

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

(c) student

course_id	*sec_id*	*semester*	*year*	*building*	*rn*	*time_id*
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

(d) section

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

(e) teaches

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

(f) takes

dept_name	*building*	*budget*
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00
Finance	Painter	120000.00
History	Painter	50000.00
Music	Packard	80000.00
Physics	Watson	70000.00

(g) department

building	*room_number*	*capacity*
Packard	101	500
Painter	514	10
Taylor	3128	70
Watson	100	30
Watson	120	50

(h) classroom